Set-quotients from an idempotent: the fixed points of an idempotent retraction satisfy the universal property of the quotient

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'lem:quotient-when-canonical-representatives' (source lines 1321-1338, Lemma 6.10.8); the proof environment starts two lines later, after \end{lem} and one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in paragraph, which supplies the definition of an idempotent function that the statement uses.

Editorial note. The retained counter values (chapter 6, section 10, theorem offset 7) make the retained lemma print as Lemma 6.10.8, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference; the numbered display (6.10.?) of the statement is defined and used inside the retained text.

Editorial scope. One original result of the HoTT Book, counted once: set-quotients presented by an idempotent, Lemma 6.10.8.

Allowed standard prerequisites (editorial).

- Universal property of set-quotients. For a set B, precomposition with q : A -> A/R is an equivalence between A/R -> B and the type of R-constant functions on A.. The statement of the counted lemma is the assertion that the type of fixed points of r satisfies this same universal property, and the proof verifies it by exhibiting the equivalence of equation (6.10.?) directly.
- Paths in a sigma type over a set. For a set A, two elements (x,p) and (y,q) of Sigma x:A (r x = x) are equal as soon as x = y, and p and q need not be compared.. It is what makes the comparisons at source lines 1343, 1356 and 1361 harmless, since B is a set and the second components are propositions.

The following lemma says that when this sort of thing happens, we don't need either general construction of quotients. (A function $r : A \to A$ is called **idempotent** if $r \circ r = r$.)

Lemma 6.10.8. *Suppose $\sim$ is a relation on a set A, and there exists an idempotent $r : A \to A$ such that $(r(x) = r(y)) \simeq (x \sim y)$ for all $x, y : A$. (This implies $\sim$ is an equivalence relation.) Then the type*

$$(A/\sim) :\equiv \Big(\sum_{x:A} r(x) = x\Big)$$

satisfies the universal property of the set-quotient of A by $\sim$, and hence is equivalent to it. In other words, there is a map $q : A \to (A/\sim)$ such that for every set B, precomposition with q induces an equivalence

$$\Big((A/\sim) \to B\Big) \simeq \Big(\sum_{(g:A\to B)} \prod_{(x,y:A)} (x \sim y) \to (g(x) = g(y))\Big). \tag{6.10.9}$$

Proof. Let $i : \prod_{(x:A)} r(r(x)) = r(x)$ witness idempotence of r. The map $q : A \to (A/\sim)$ is defined by $q(x) :\equiv (r(x), i(x))$. Note that since A is a set, we have $q(x) = q(y)$ if and only if $r(x) = r(y)$, hence (by assumption) if and only if $x \sim y$. We define a map e from left to right in (6.10.9) by

$$e(f) :\equiv (f \circ q, _),$$

where the underscore _ denotes the following proof: if $x, y : A$ and $x \sim y$, then $q(x) = q(y)$ as observed above, hence $f(q(x)) = f(q(y))$. To see that e is an equivalence, consider the map e' in the opposite direction defined by

$$e'(g, s)(x, p) :\equiv g(x).$$

Given any $f : (A/\sim) \to B$,

$$e'(e(f))(x, p) \equiv f(q(x)) \equiv f(r(x), i(x)) = f(x, p)$$

where the last equality holds because $p : r(x) = x$ and so $(x, p) = (r(x), i(x))$ because A is a set. Similarly we compute

$$e(e'(g, s)) \equiv e(g \circ \mathsf{pr}_1) \equiv (g \circ \mathsf{pr}_1 \circ q, _).$$

Because B is a set we need not worry about the _ part, while for the first component we have

$$g(\mathsf{pr}_1(q(x))) \equiv g(r(x)) = g(x),$$

where the last equation holds because $r(x) \sim x$, and g respects $\sim$ by the assumption s. □